

Sri Lanka Institute of Information Technology



BSc (Hons) in Information Technology Specializing in Data Science

IT3021 – Data Warehousing and Business Intelligence

Year 3 Semester 1

Assignment 1

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Introduction

In today's data-driven environment, organizations generate large volumes of data from different operational systems such as sales, customers, and order transactions. However, this data is often stored in separate sources and formats, making it difficult to analyze and use for decision-making. Data Warehousing and Business Intelligence (DWBI) solutions help to integrate, transform, and organize this data into a structured format that supports effective analysis and reporting.

This assignment focuses on designing and implementing a complete data warehouse solution using a sales dataset. The data is first extracted from source systems and loaded into a staging area, where it is cleaned and prepared for further processing. Then, the data is transformed and loaded into a data warehouse designed using a star schema, which includes fact and dimension tables such as DimCustomer, FactSales and other related dimensions.

The solution is developed using SQL Server Integration Services (SSIS), where ETL (Extract, Transform, Load) processes are created to handle data movement, data cleansing, lookup operations, and surrogate key generation. Additional transformations such as derived columns and data type conversions are also applied to improve data quality and consistency.

Through this assignment, key data warehousing concepts are practically implemented, including staging design, dimension and fact table creation, and ETL workflow development. The final data warehouse enables efficient querying and analysis of sales data, helping to identify trends, customer behavior, and business performance over time.

1. Data Set Selection

1.1 Introduction

For this assignment, a Retail Sales Transaction System dataset was selected as the OLTP source. This dataset represents real-world business operations related to sales, customers, products, and orders. It includes transactional data such as order details, shipping information, customer records, and sales performance.

It was specifically developed for this assignment and has not been used in any lab sessions, nor does it use any OLAP-style fact/dimension structures or the AdventureWorks dataset.

The dataset is suitable for demonstrating data warehousing concepts because it reflects how data is typically stored in operational systems before being transformed into an analytical structure.

This dataset provides a practical business scenario, making it ideal for implementing ETL processes, dimensional modeling, and business intelligence solutions

1.2 Dataset Overview

This dataset contains 100,000 records of retail store sales data from various outlets across India over the past 5 years (2019-2023). It includes information about customer purchases, product categories, sales, discounts, and profit margins across different regions and outlet types.

Customer: Stores customer details such as Customer ID, Name, and demographic information.

Orders: Contains transaction-level data including Order ID, Sales amount, and dates.

Product: Represents product categories and types of goods sold.

This variety of attributes allows for multiple analytical perspectives such as **sales trends, customer segmentation, and profitability analysis.**

1.3 Source Types and Formats

To simulate a real-world enterprise data environment, the dataset was intentionally divided into multiple heterogeneous data sources:

Source Type	Example Files/Tables	Description
SQL Database	DWBI_Assignment DB Dbo.orders	Sales transactions
Excel file	products.xlsx	Product information
CSV file	customer.csv	Customer Details

1.4 Coverage and Realism

The dataset provides a high level of realism due to the following characteristics:

- Covers multiple geographic regions within India
- Includes different customer segments
- Contains various product categories
- Supports multiple shipping methods
- Includes financial measures such as sales, discounts, and profit

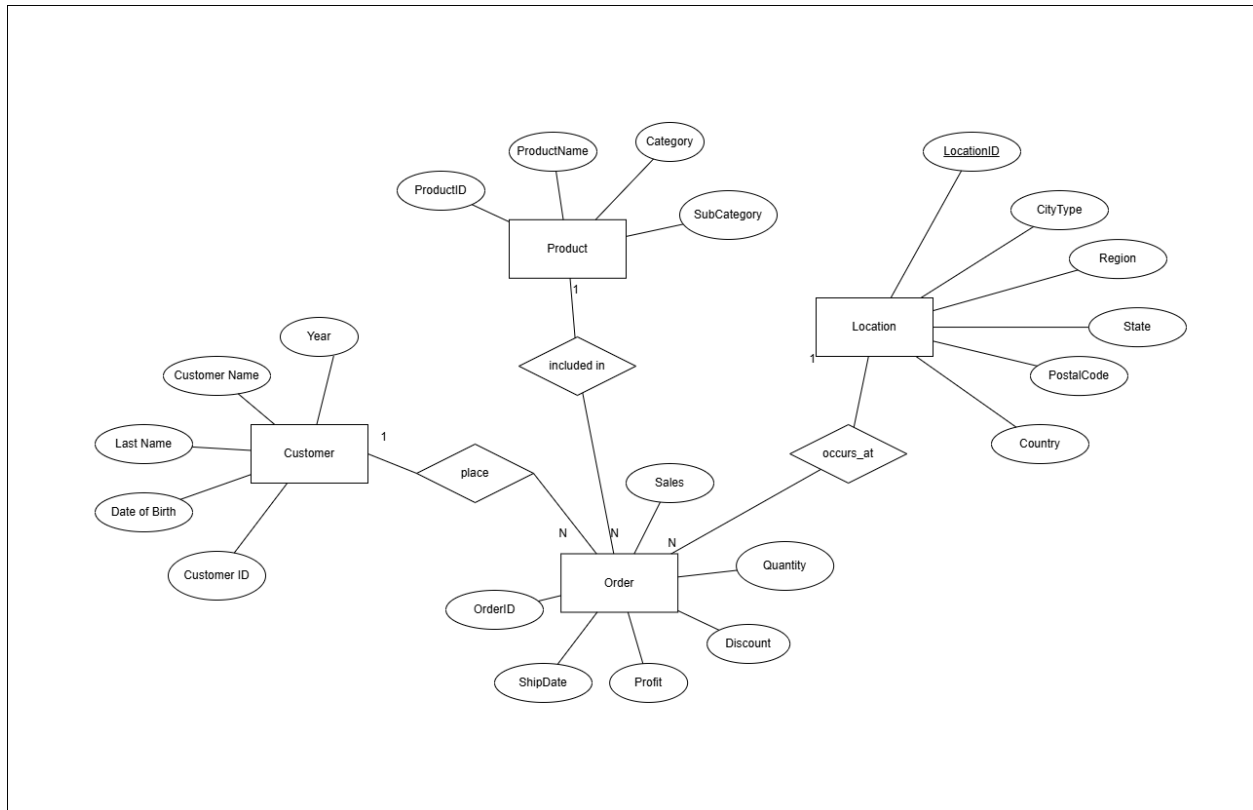
These features make the dataset suitable for implementing a business intelligence solution that reflects real-world retail analytics.

1.5 Suitability for Data Warehousing and BI

The dataset satisfies all major requirements specified in the assignment:

Requirement	Details
OLTP structure	Normalized database with transactional records
Novel Scenario	
One year of data	Covers 5 years of data
Multiple source types	SQL Server database + external TXT file
Complex ETL support	
Warehouse-ready dimensional modeling	Supports star schema with fact and multiple dimension tables
SSAS cube and reporting compatibility	Hierarchies available

1.6 Entity–Relationship Diagram of the Source Dataset



2. Source Data Description

To simulate a realistic enterprise data warehousing environment, the source data for this project was obtained from **multiple heterogeneous data sources**, including a structured SQL Server database and external files (CSV and Excel). This approach demonstrates how ETL processes integrate data from different formats into a unified data warehouse.

2.1 SQL Server Database Source

The primary structured data source used in this project is a SQL Server database named:

DWBI_Assignment

This database represents a transactional (OLTP-style) system containing sales-related data. The main table used from this database is:

- **StgOrders** – Stores transactional sales data for the retail store.

This table contains key business information, including:

- Order ID – Unique identifier for each transaction
- Customer ID – Identifies the customer who made the purchase
- Product ID – Identifies the purchased product
- Order Date – Date when the order was placed
- Ship Date – Date when the order was shipped
- Sales – Total sales amount
- Quantity – Number of items purchased
- Discount – Discount applied to the order
- Profit – Profit earned from the transaction

This data represents the **core transactional operations of a retail business**, similar to a real-world sales system.

Although the original dataset was not fully normalized like a traditional OLTP database, it was structured and stored in SQL Server to simulate a relational source system. Data from this table was extracted using **OLE DB Source components in SSIS** and loaded into staging tables for further transformation.

2.2 File-Based Sources

In addition to the SQL Server database, external file-based sources were used to simulate integration from non-database systems.

2.2.1 Customer Data (CSV File)

A CSV file named:

customer.csv

was used to store customer-related information. This file contains the following fields:

- Customer ID
- Customer Name
- Last Name
- Date of Birth
- Segment

This file represents data coming from an external system such as a customer management platform.

2.2.2 Product Data (Excel File)

An Excel file named:

product.xlsx

was used to store product-related information. It includes:

- Product ID
- Product Name
- Category
- Sub-Category

This simulates data coming from a **product inventory or catalog system**.

The CSV and Excel files used in this project are structured using standard data formats, where the CSV file stores data as comma-separated values and the Excel file organizes data in a tabular worksheet format. These external data sources were integrated into the ETL pipeline using a Flat File Connection Manager for the CSV file and an Excel Connection Manager for the Excel file within SQL Server Integration Services.

During the ETL process, data was first extracted from each source and then loaded into staging tables such as StgCustomer, StgProduct, and StgOrders. After staging, the data underwent necessary transformations and cleansing operations before being loaded into the data warehouse.

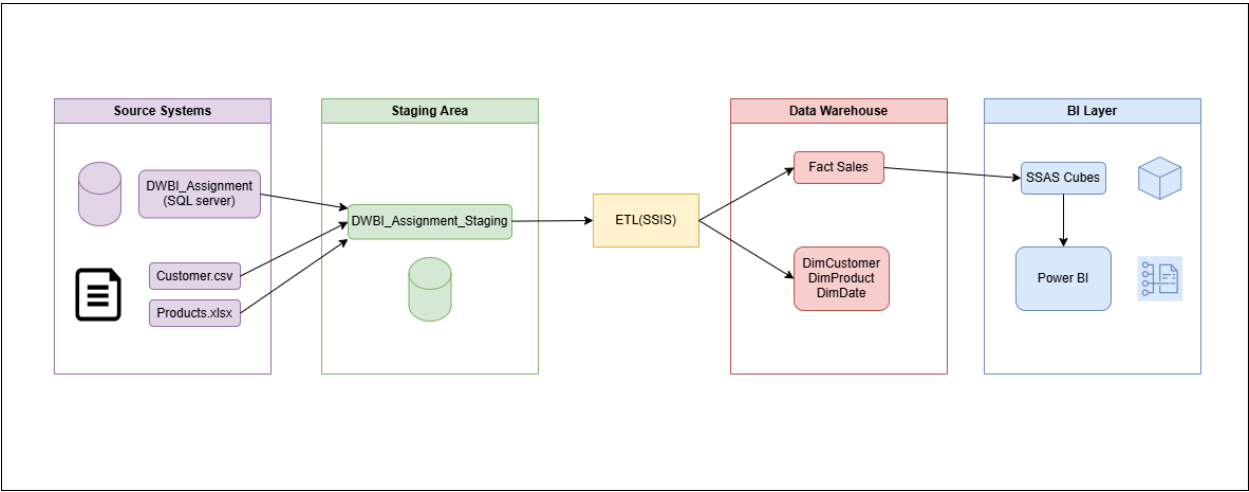
The use of multiple data sources in this project highlights important real-world challenges such as integrating data from different systems, handling diverse data formats including CSV, Excel, and SQL, and ensuring data consistency and quality during transformation. This approach demonstrates the flexibility and capability of ETL tools in consolidating heterogeneous data into a single, structured, and reliable data warehouse for analytical purposes.

3. Solution Architecture

A high-level Data Warehouse (DW) and Business Intelligence (BI) solution architecture was designed to support the integration, transformation, and analysis of sales transaction data. The architecture reflects a realistic enterprise BI pipeline with multiple layers, starting from operational data sources and ending with user-facing analytical tools.

3.1 Architectural Diagram

The following diagram illustrates the key components of the DW & BI solution implemented for the retail store sales.



3.2 Component Summary

Layer	Component	Description
Source Systems	customer.csv (CSV File)	Contains customer-related data such as Customer ID, Name, Last Name, Date of Birth, and Segment. This simulates data coming from an external customer management system.
	product.xlsx (Excel File)	Stores product details including Product ID, Product Name, Category, and Sub-Category. This represents a product catalog or inventory system.
	DWBI_Assignment (SQL Server – Orders Table)	Contains transactional sales data such as Order ID, Customer ID, Product ID, Order Date, Ship Date, Sales, Quantity, Discount, and Profit. This acts as the core OLTP-like system for sales operations.
Staging Area	StgCustomer StgProduct StgOrders	Temporary tables used to store raw data extracted from different sources. This layer is used for initial data loading, cleansing, and preparation before transformation into the data warehouse.
ETL Process	SQL Server Integration Services (SSIS Packages)	Responsible for extracting data from multiple sources, transforming it using Data Conversion, Lookup, Conditional Split, and Derived Column transformations, and loading it into dimension and fact tables.
	as_Load_Staging.dtsx	Extracts data from CSV, Excel, and SQL sources and loads it into staging tables. Ensures data is ready for transformation.
	as_Load_DW.dtsx	Transforms staging data and loads it into the data warehouse. Includes surrogate key generation, lookup transformations, and loading of dimensions and fact tables.
	Update_FactSales.dtsx	Updates the accumulating fact table by setting accm_txn_complete_time and calculating

		txn_process_time_hours based on transaction completion data.
Data Warehouse	DWBI_Assignment_DW	A dimensional data warehouse implemented using a star schema design. It integrates cleaned and transformed data for analytical purposes.
	DimCustomer	Stores customer-related attributes with surrogate key CustomerSK and business key AlternateCustomerID.
	DimProduct	Stores product-related attributes with surrogate key ProductSK and business key AlternateProductID.
	DimDate	Stores date-related attributes such as Year, Month, and Day, enabling time-based analysis.
	FactSales	Central fact table that stores transactional measures such as Sales, Quantity, Discount, and Profit, along with foreign keys referencing dimension tables. It also includes accumulating fact fields such as transaction timestamps.
BI Layer	Reporting / Analytics (Future Scope)	The data warehouse is designed to support analytical tools such as dashboards and reporting systems. Although not implemented, it is compatible with tools like Power BI for visualization and reporting.

4. Data Warehouse Design & Development

A **star schema-based dimensional model** was designed for the retail sales dataset to support efficient analytical processing and reporting. This model organizes data into a central fact table surrounded by multiple dimension tables, enabling fast query performance and simplified data analysis.

The data warehouse is structured to analyze sales performance across different dimensions such as customers, products, and time. It supports key business insights such as sales trends, customer behavior, product performance, and profitability analysis. The schema consists of one central fact table (FactSales) and three primary dimension tables (DimCustomer, DimProduct, and DimDate), fulfilling the assignments requirements

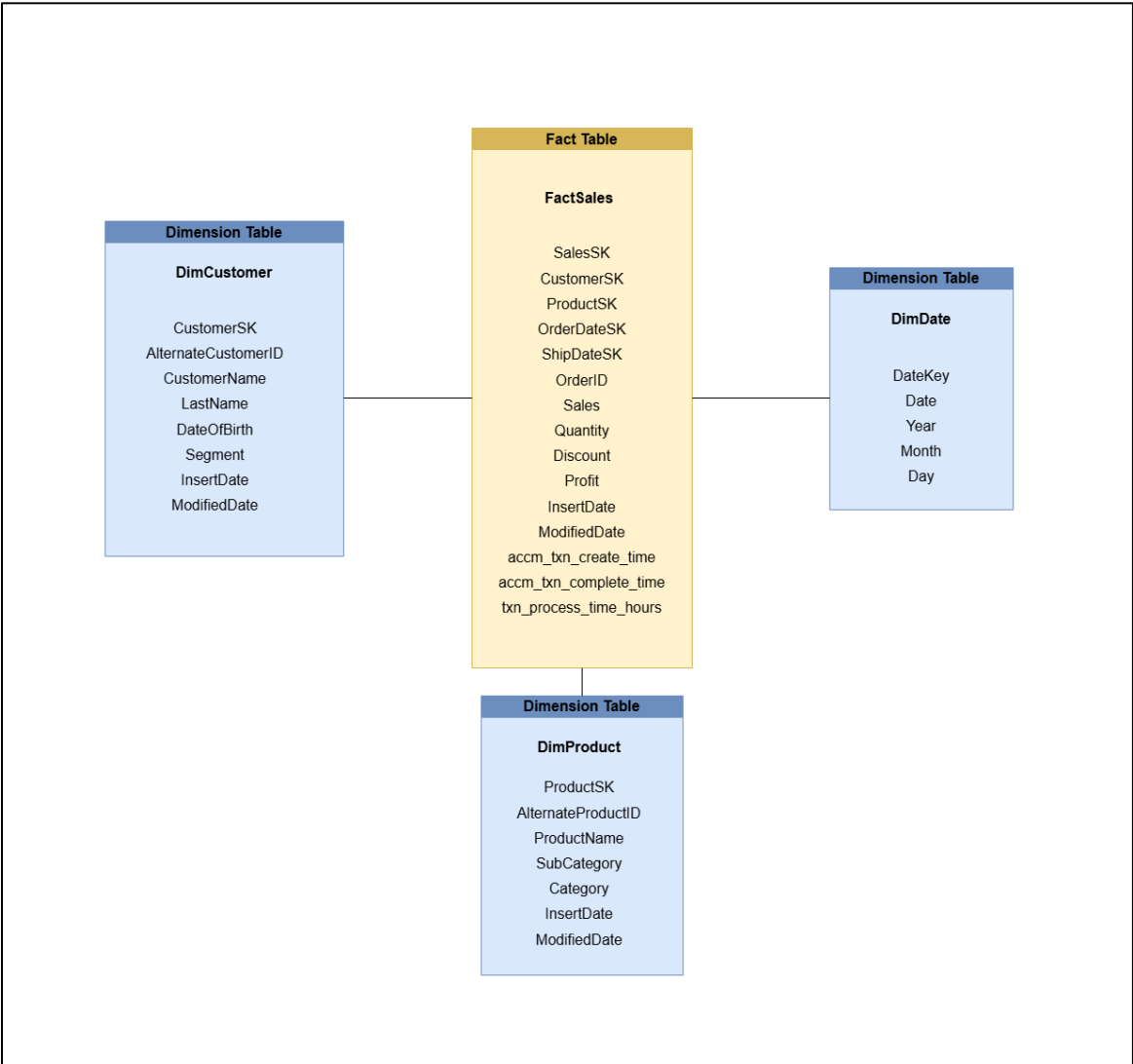
4.1 Dimensional Model Description

Table Name	Type	Description
FactSales	Fact Table	Stores transactional sales data for each order. It includes measures such as Sales, Quantity, Discount, and Profit. It also contains foreign keys referencing dimension tables and additional accumulating fact fields such as transaction creation time, completion time, and processing duration.
DimCustomer	Dimension	Contains customer-related attributes such as Customer Name, Last Name, and Segment. A surrogate key (CustomerSK) is used as the primary key, while AlternateCustomerID acts as the natural/business key. This table supports customer segmentation and analysis.
DimProduct	Dimension	Stores product-related details including Product Name, Category, and Sub-Category. It uses a surrogate key (ProductSK) and a natural key (AlternateProductID) to uniquely identify products. This dimension enables product-level analysis and category-based reporting.
DimDate	Dimension	A time-based dimension that stores date attributes such as Date, Year, Month, and Day. It uses an integer key (DateKey) formatted as YYYYMMDD. This dimension

		allows time-based analysis such as trends, seasonality, and comparisons across periods.
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4.2 Relational Diagram

The star schema



4.3 Design Assumptions

- A star schema was selected due to its simplicity and high performance in analytical queries.
- Each row in FactSales represents a single sales transaction (Order level).
- Surrogate keys (CustomerSK, ProductSK, SalesSK) are used instead of natural keys to improve join performance and ensure consistency.
- The OrderID is used as a natural key to uniquely identify transactions and support updates in the accumulating fact table (Task 6).
- The DimDate table is used as a shared (conformed) dimension for both Order Date and Ship Date.
- InsertDate and ModifiedDate fields are used to track data loading and updates in the warehouse.
- Accumulating fact columns are used to analyze transaction lifecycle and processing time.

5. ETL Development

To populate the data warehouse with clean and integrated data, three ETL packages were developed using SQL Server Integration Services (SSIS). The ETL process is divided into three main stages:

1. Data Extraction to Staging
2. Transformation and Loading to the Data Warehouse
3. Updating Accumulating Fact Table Attributes

Each package demonstrates key SSIS tasks such as connection setup, control and data flow design, transformations, and load operations from multiple data sources.

5.1 SSIS Packages Overview

Four SSIS packages were created to organize the ETL workflow logically and modularly:

Package Name	Purpose
as_Load_Staging.dtsx	Extracts data from OLTP and flat file sources into staging tables.
as_Load_DW.dtsx	Transforms and loads data into dimension and fact tables.
Update_FactSales.dtsx	Updates accumulating fact attributes using staged completion data.
DataProfiling.dtsx	Used to assess the quality and structure of source data.

This separation ensures that each stage of the ETL pipeline can be executed, tested, and maintained independently.

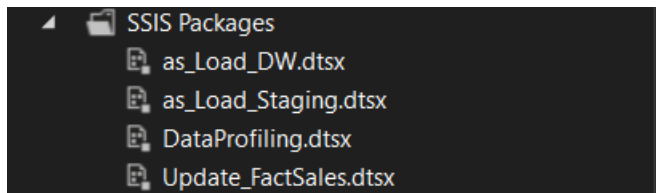


Figure 5.1: SSIS package modules as seen in Solution Explorer.

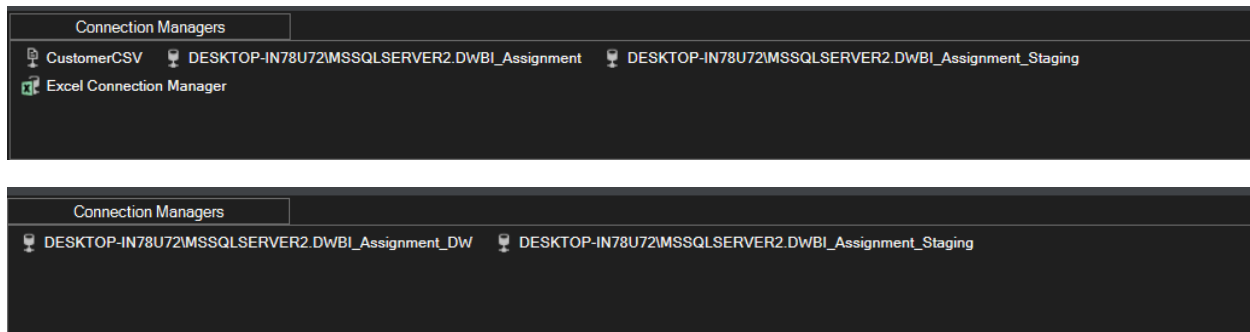
5.2 Connection Managers

All ETL packages utilize multiple connection types to handle data from both databases and text files.

The following connections were configured:

- SQL Server – OLE DB Connections:
 - DWBI_Assignment_DW (data warehouse)
 - DWBI_Assignment_Staging (staging database)
 - DWBI_Assignment (data warehouse)
- Flat File Connections:
 - CustomerCSV
 - Excel Connection Manager

These connection managers enable seamless integration between heterogeneous data sources.



5.3 Package 1: as_Load_Staging.dtsx

This package is responsible for extracting raw data from multiple sources and loading it into staging tables.

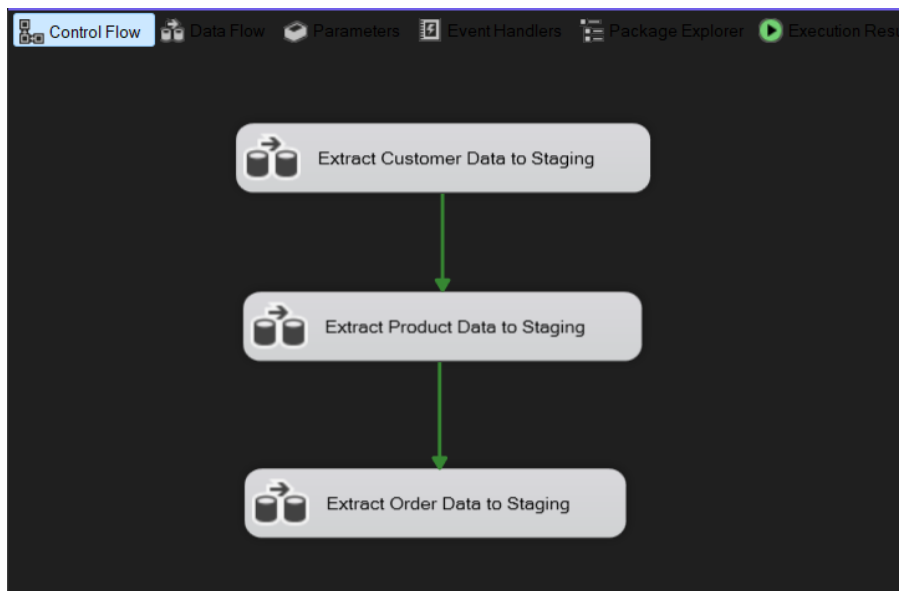
Data was extracted from:

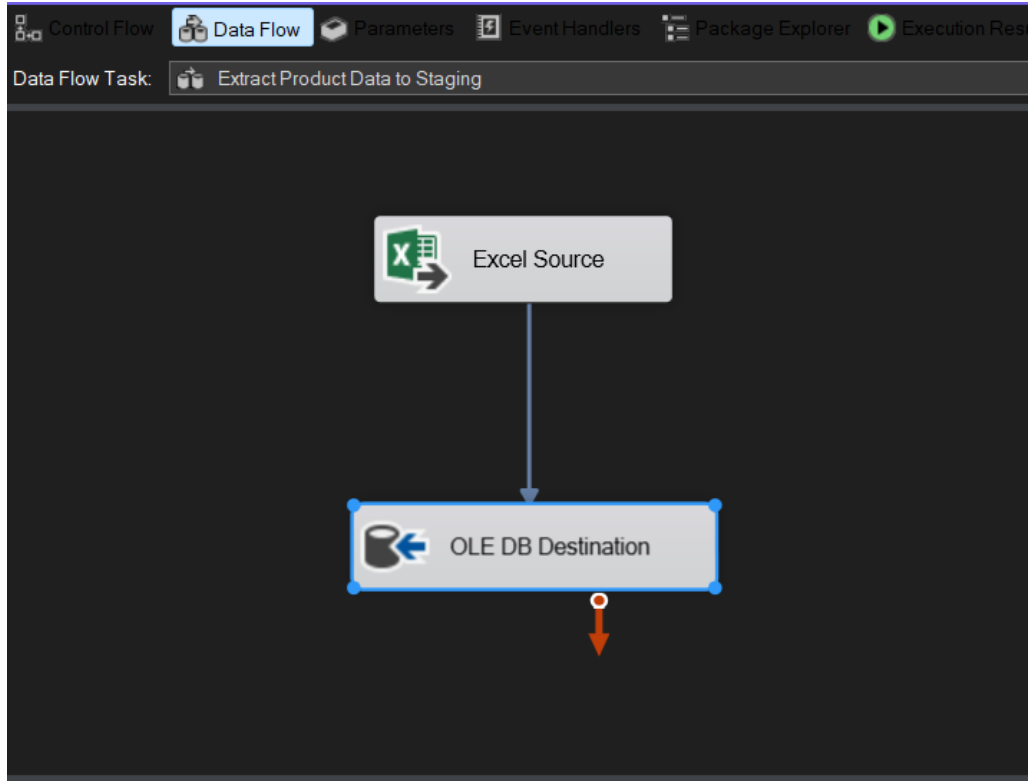
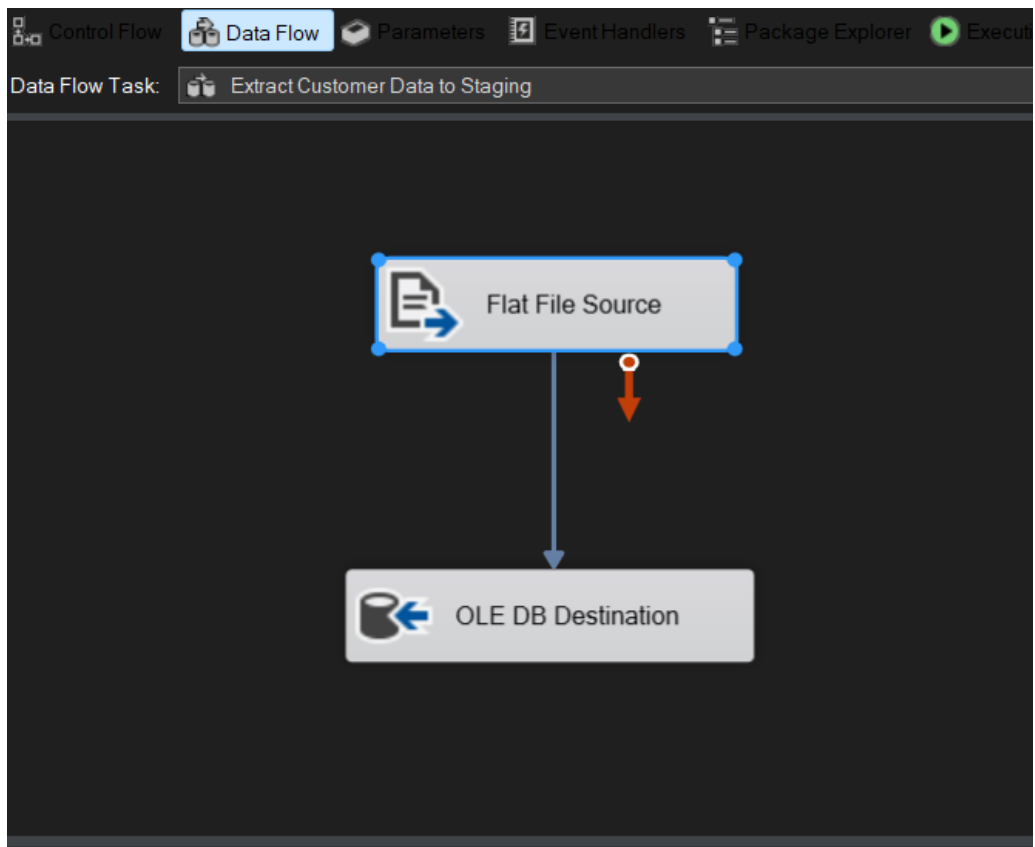
- CSV file (customer data)
- Excel file (product data)
- SQL Server table (orders data)

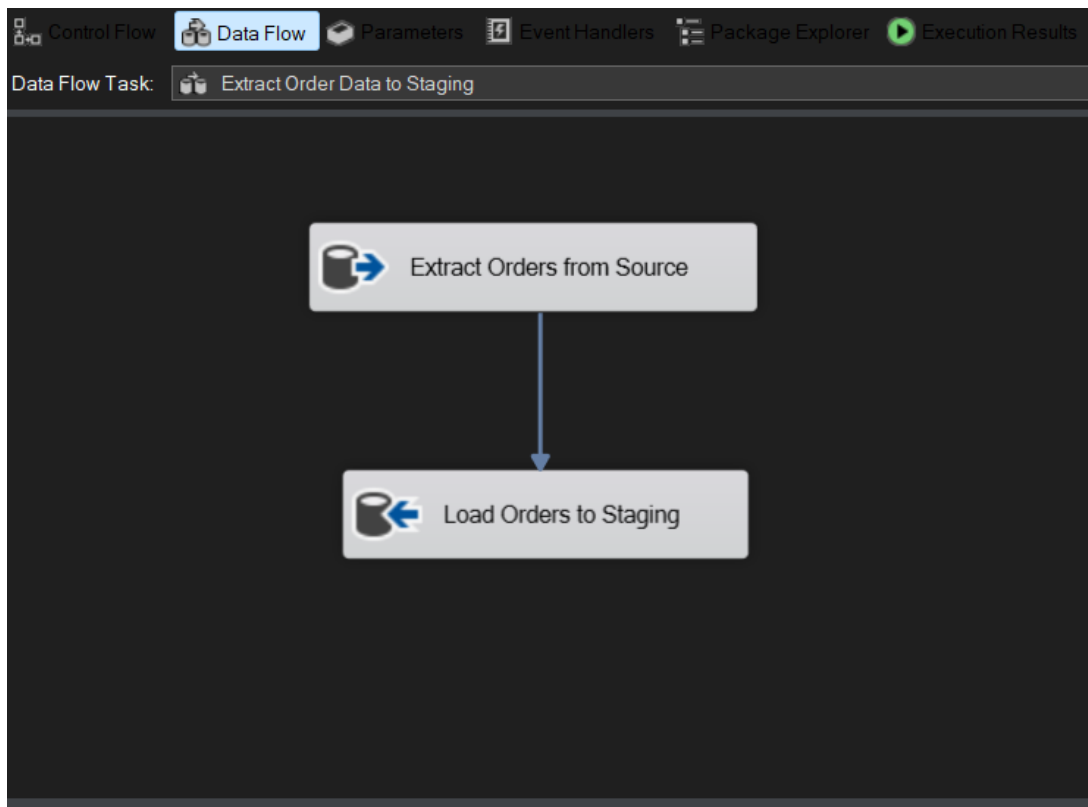
Each data flow uses a simple structure:

- **Source → Data Conversion → OLE DB Destination**

Before loading data, staging tables were truncated to avoid duplicate records and ensure consistent data loading. This approach ensures that staging always contains fresh data for further processing.

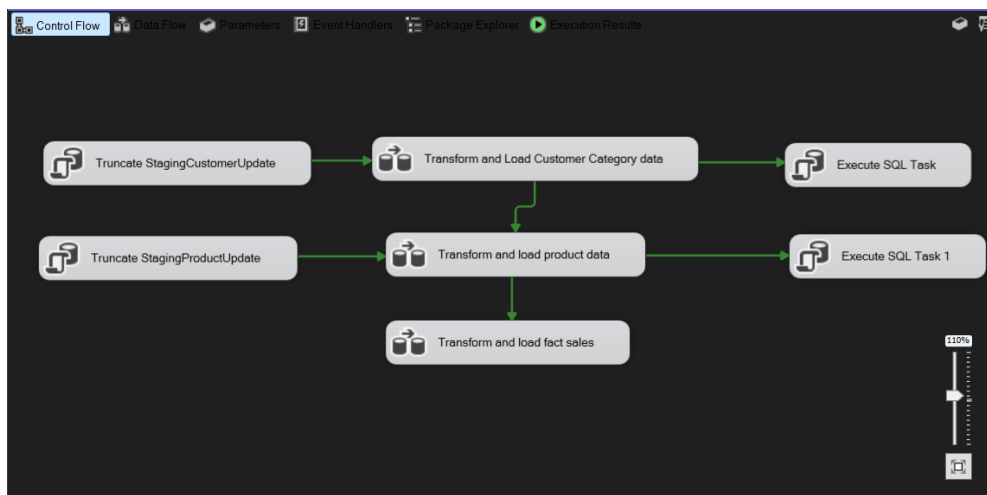






5.4 Package 2: as_Load_DW.dtsx

This package handles the transformation and loading of clean data into the dimensional and fact tables of the data warehouse.



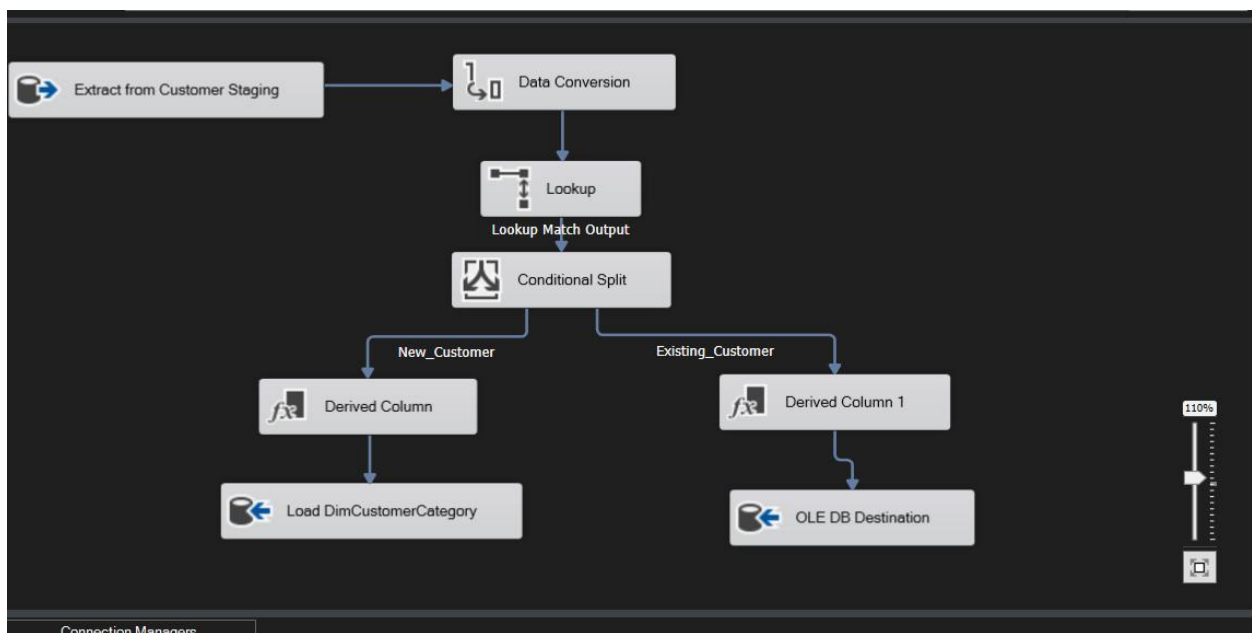
5.4.1 Data Flow – Load DimCustomer

The DimCustomer loading process begins by extracting data from the staging table. Data types are first standardized using a Data Conversion transformation to ensure compatibility with the data warehouse schema.

A Lookup transformation is then used to compare incoming records with existing records in the DimCustomer table using the natural key (AlternateCustomerID). Based on this comparison, a Conditional Split transformation separates the data into two streams:

- **New Customers** → Inserted into DimCustomer
- **Existing Customers** → Updated using a staging update table

Derived Column transformations are used to populate metadata fields such as InsertDate and ModifiedDate. This ensures proper tracking of data loading and updates.



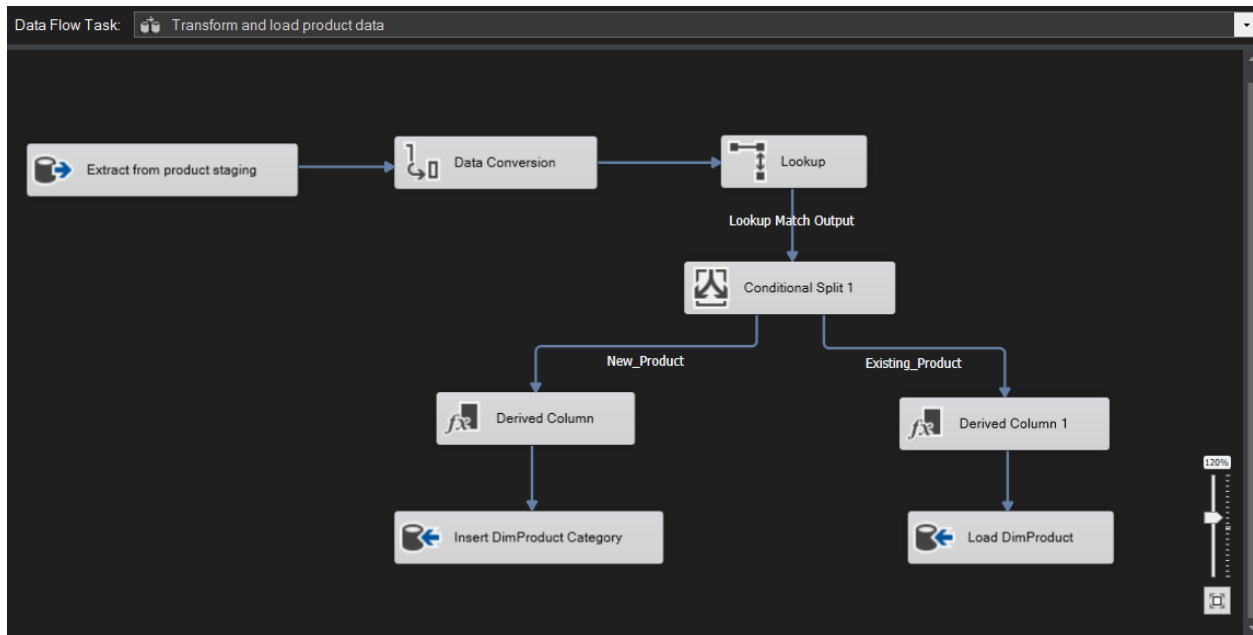
5.4.2 Data Flow – Load DimProduct

The DimProduct ETL process follows a similar pattern to DimCustomer. Data is extracted from the staging table, converted into appropriate data types, and then matched against existing records using a Lookup transformation.

A Conditional Split separates new and existing products:

- New products are inserted into the dimension table
- Existing products are updated using a staging update mechanism

This approach ensures that the dimension table remains consistent and up to date.



5.4.3 Data Flow – Load FactSales

The FactSales data flow is more complex as it integrates multiple dimensions.

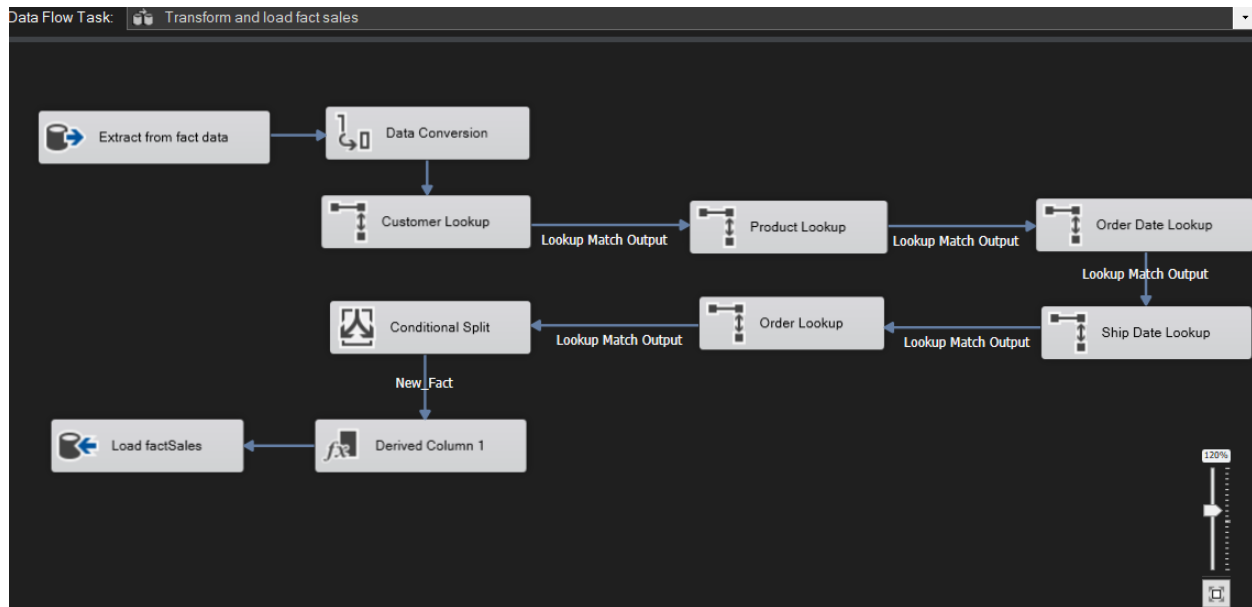
The process includes the following steps:

1. Data is extracted from the StgOrders table
2. Data Conversion is applied to ensure compatibility with the warehouse schema
3. Multiple Lookup transformations are performed to retrieve surrogate keys:
 - Customer Lookup → CustomerSK
 - Product Lookup → ProductSK
 - Date Lookups → OrderDateSK and ShipDateSK
4. A Derived Column transformation is used to generate:
 - InsertDate

- ModifiedDate
- accm_txn_create_time (set to current system date)

5. Data is then loaded into the FactSales table using an OLE DB Destination

This ensures that all fact records are properly linked to their corresponding dimensions through surrogate keys.



5.5 Package 3: UpdateFactEnrollment.dtsx

This package implements the Accumulating Fact Table logic, which updates existing records in the FactSales table when new information becomes available.

1.Prepare Update Dataset

A separate SQL table named FactSales_Update was created. This table contains:

- OrderID (Natural Key)
- accm_txn_complete_time

This dataset simulates the completion of transactions after a certain period.

2. Lookup Transformation

A Lookup transformation is used to match records from the update table with the FactSales table using OrderID.

This step ensures that the correct fact records are identified for updating.

3. Derived Column Transformation

A Derived Column transformation is used to calculate:

- txn_process_time_hours

This is calculated as the difference between:

- accm_txn_complete_time
- accm_txn_create_time

using the DATEDIFF function.

4. Update Fact Table

An update operation is performed using SQL:

```
UPDATE FactSales
SET accm_txn_complete_time = ?,
    txn_process_time_hours = ?
WHERE OrderID = ?
```

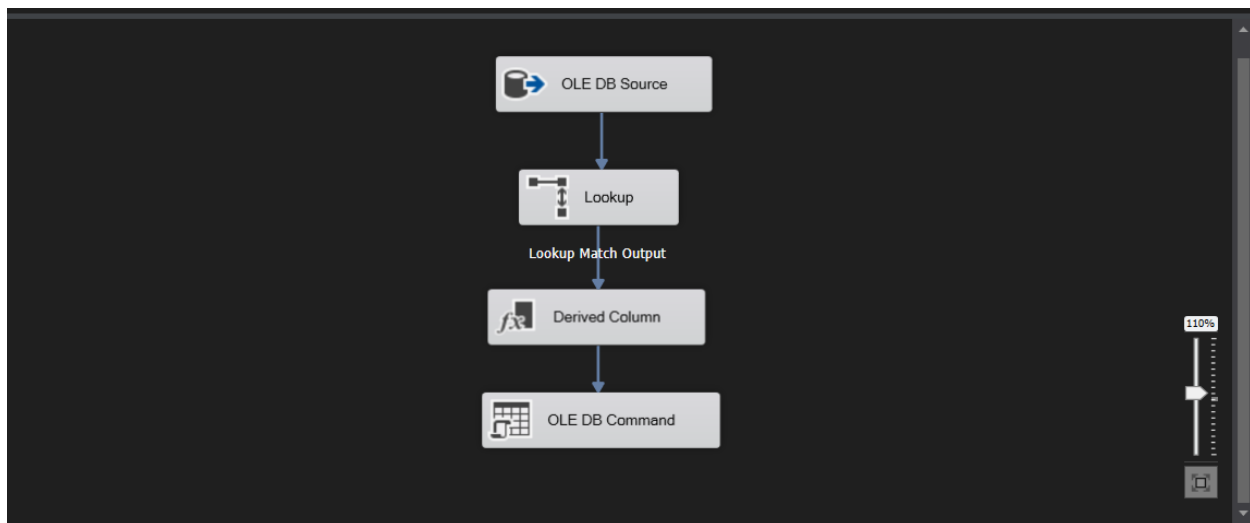
This updates the existing records in the fact table with completion time and processing duration.

The OrderID was used as the natural key for updates instead of SalesSK because:

- Surrogate keys (SalesSK) may change during reloads
- Natural keys remain stable and consistent across systems

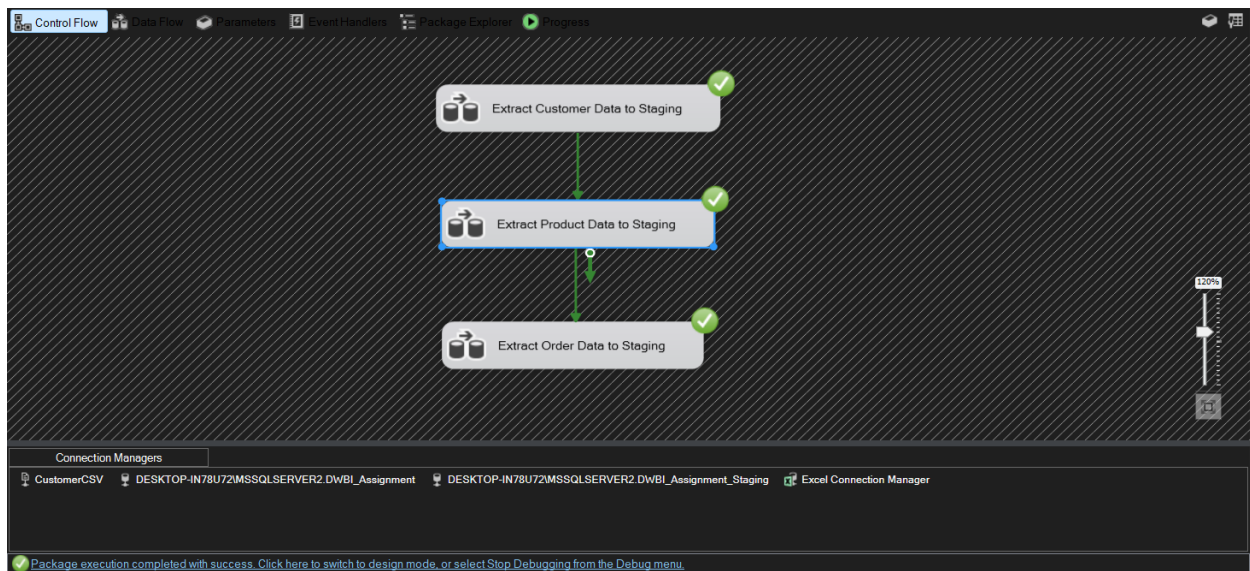
This ensures reliable and accurate updates in the accumulating fact table.

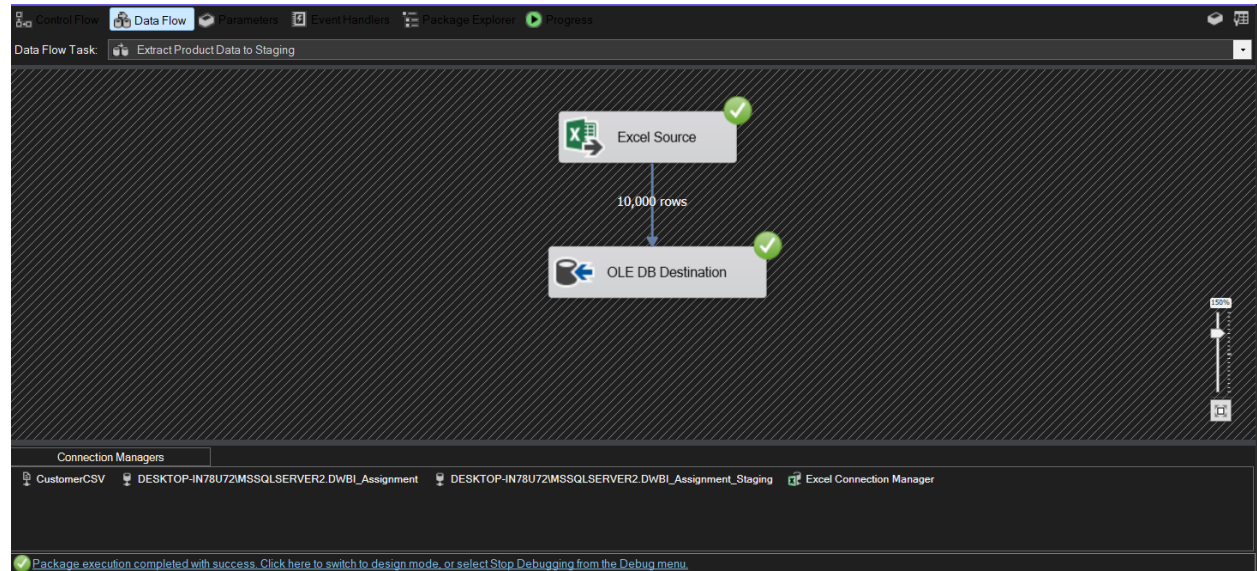
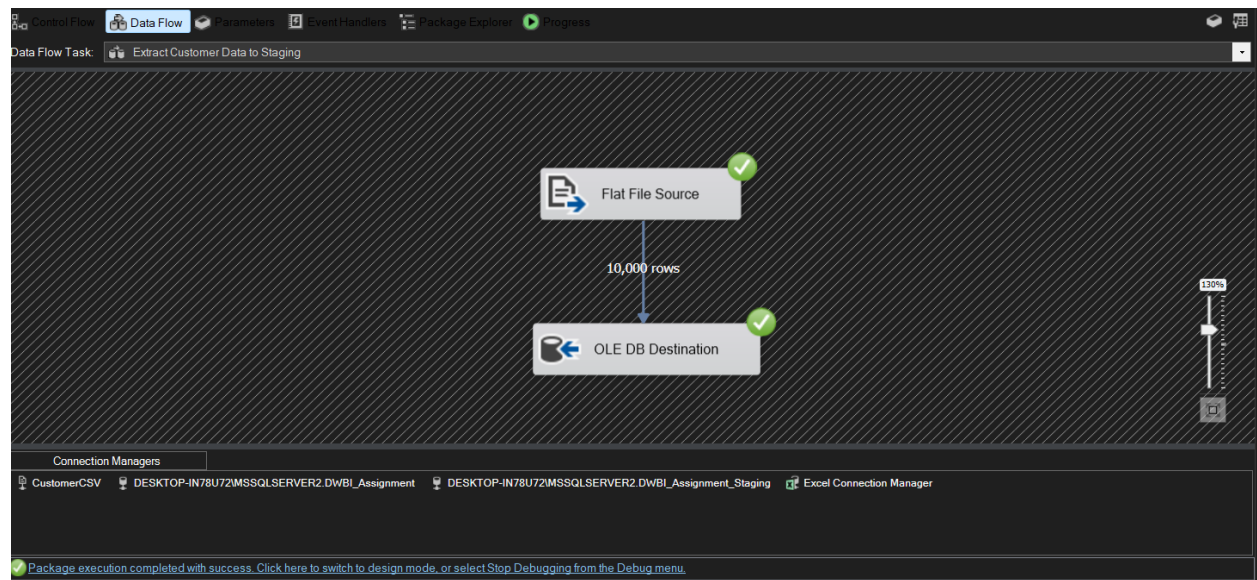


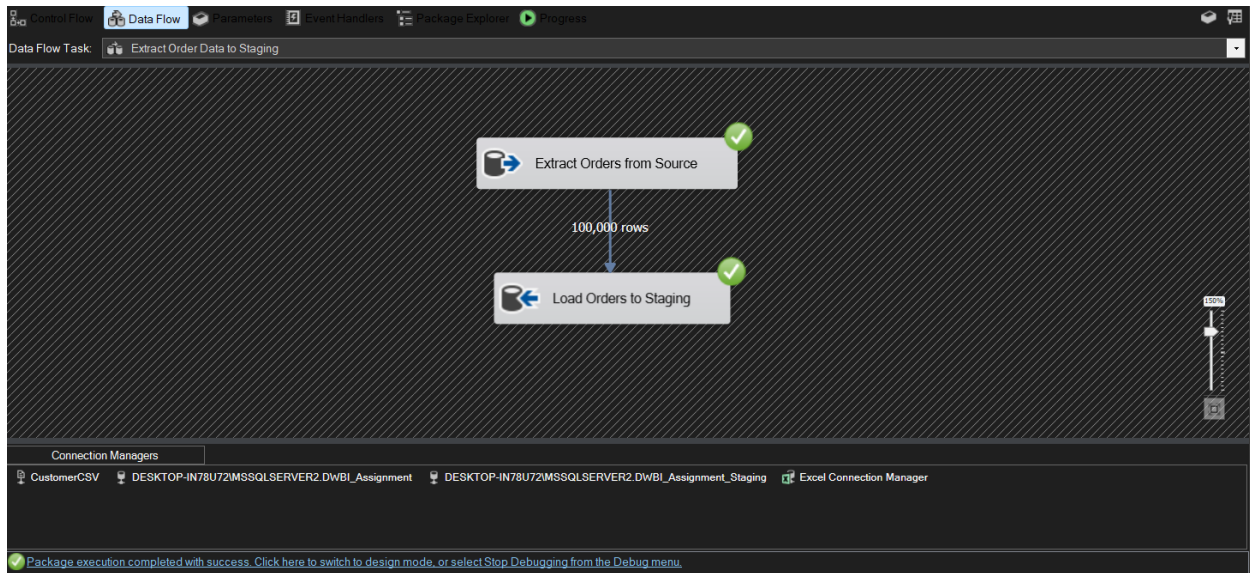


Evidence of Row Counts: Data Load into Staging Tables

To validate that all source data was correctly extracted and loaded into the staging layer, screenshots from the as_Load_Staging.dtsx SSIS package execution are provided. Each screenshot includes the number of rows successfully loaded into its respective staging table.

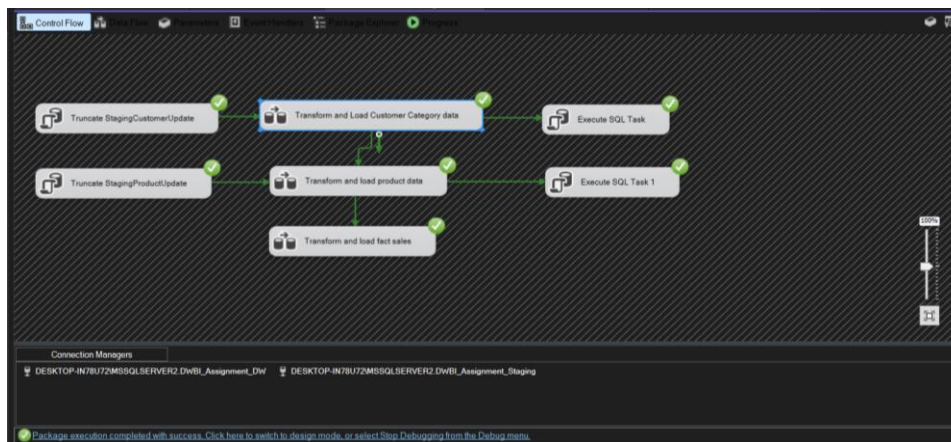


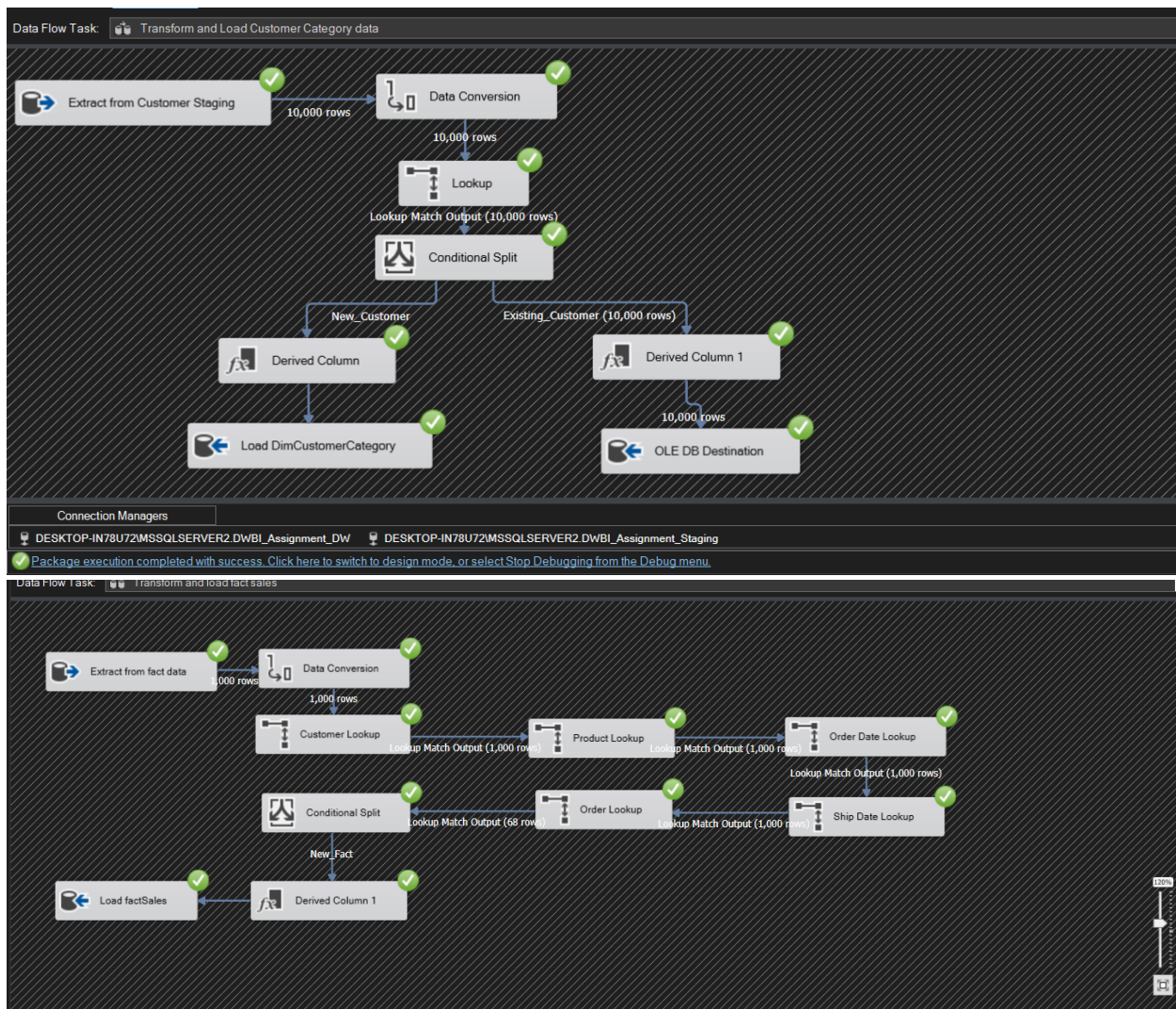




Evidence of Row Counts: Data Load into Data Warehouse Tables

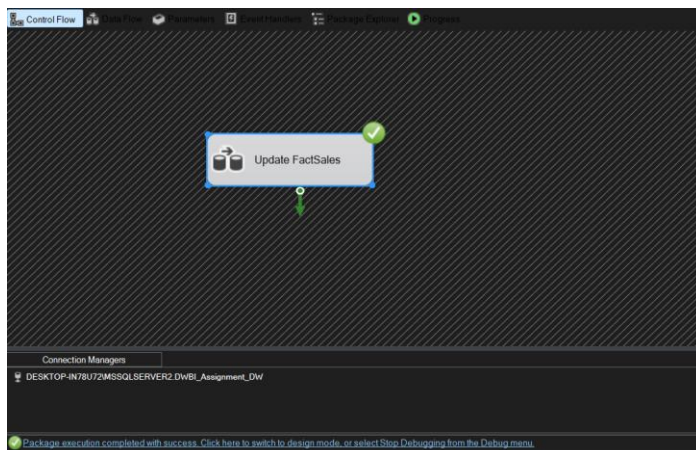
To verify the transformation and successful loading of data into the dimensional tables, screenshots from the as_Load_DW.dtsx SSIS package are provided below. These highlight the number of rows inserted into each dimension after transformation, key resolution, and business logic handling

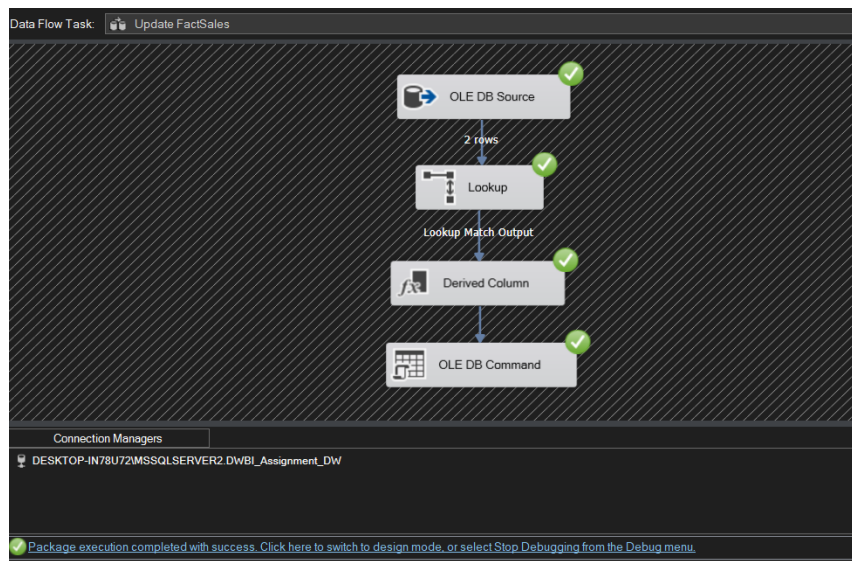




Evidence of Row Counts: Staging Completion Data from Flat File The

following figure captures the successful data flow execution of the UpdateFactSales.dtsx package's staging task.





6. Conclusion

This project successfully demonstrated the complete implementation of a data warehouse solution for retail sales analysis.

The solution integrated data from multiple heterogeneous sources, including CSV files, Excel files, and SQL Server tables. A star schema-based dimensional model was designed to support efficient analytical queries.

Key data warehousing concepts such as surrogate key generation, lookup transformations, and Slowly Changing Dimensions (Type 1) were implemented effectively. Additionally, accumulating fact table logic was developed to track transaction lifecycle and processing time.

The ETL process was implemented using SSIS in a modular and structured manner, ensuring maintainability and scalability. The final data warehouse is capable of supporting advanced analytics and can be easily integrated with business intelligence tools such as Power BI for reporting and visualization.

Overall, this assignment provided practical experience in data integration, transformation, and warehouse design, demonstrating the importance of ETL processes in building reliable business intelligence systems.